NIDAR RescueSwarm — Bill of Materials

Three-aircraft programme · Revision 2026-09-20 · all prices INR

Parts, three aircraft	**4,93,978**	Parts per aircraft	**1,37,509**
Released (tax where owed + 15 % contingency)	**5,71,171**	Line items	**61**

Architecture

Flight controller	Holybro Kakute H7 V1.5 (ArduPilot), stacked on a 4-in-1 ESC
Companion	Raspberry Pi 5 4 GB + AI HAT+ 26 TOPS; Arducam 12.3 MP, 6 mm lens
Positioning	ZED-F9P RTK on every aircraft (GenX F9P Air, compass built in); local F9P base, RTCM over E22 LoRa, no NTRIP
Video	Digital, wfb-ng on Pi OS (BL-M8812EU2), CV overlay on the feed; one ground receiver per aircraft
Links	ELRS 2.4 GHz safety pilot (TX12 MKII); E22 868 MHz primarily for RTCM
Power	6S 16 Ah HV LiPo; ESC current sensor to FC; 8 A UBEC for Pi, 3 A UBEC for video radio, separate servo UBEC

Summary by group

Group	Lines	Per aircraft	Programme
Test instruments	6	–	3,246
Avionics, sensing and RTK	22	72,317	2,44,677
Airframe, propulsion and power	19	65,192	2,03,243
Flight-test support	7	–	29,867
Ground station	7	–	12,945
Total parts	**61**	**1,37,509**	**4,93,978**

Line items

Item	Part	/AC	Qty	Unit	Total	Source
Test instruments						**3,246**
Load-cell amplifier	7Semi HX711 24-bit ADC	–	1	104	104	Robu
Load cell	REES52 20 kg load cell + HX711 breakout	–	1	854	854	Amazon India
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	–	1	373	373	Flipkart
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	–	1	1,140	1,140	OnlyScrews
Threadlocker	Loctite 243, 50 ml	–	1	664	664	Amazon India
Throttle source	Servo tester / ESC consistency tester, PPM	–	1	111	111	Robokits
Avionics, sensing and RTK						**2,44,677**
Flight controller	Holybro Kakute H7 V1.5, no Bluetooth	1	3	9,269	27,807	Indian Robo Store
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	3	161	483	Robu
Autopilot log card	SanDisk 32 GB microSDHC, class 10	1	3	1,549	4,647	Robocraze
Companion computer	Raspberry Pi 5, 4 GB	1	3	11,049	33,147	Yantronix
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	3	11,749	35,247	Robu
Compute cooling	Official Raspberry Pi 5 active cooler	1	3	519	1,557	Robu
Storage	SanDisk High Endurance 128 GB microSD	1	3	3,989	11,967	Flipkart
GNSS RTK rover	GenX F9P Air: ZED-F9P L1/L2 + helix + QMC5883 compass, ∼20 g	1	3	15,392	46,176	Robokits
GNSS RTK base	SmartElex ZED-F9P-04B RTK board, U.FL	–	1	17,965	17,965	Robocraze
GNSS antenna, base	LS-125-A L1/L2/L5 survey antenna	–	1	8,123	8,123	Robokits
GNSS feeder, base	U.FL to SMA female pigtail, 10 cm	–	1	49	49	Robocraze
GNSS cable, base	RG58 50 ohm, 3 m, TNC-M to SMA-M, made locally	–	1	600	600	local
RTK base radio	EByte E22-900T22D-V2, SX1262, 22 dBm	–	1	989	989	Robu
Height rangefinder	Benewake TFmini-S, 12 m lidar, UART	1	3	2,827	8,481	Indian Robo Store
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	3	6,799	20,397	Robu
Video transmitter	LB-Link BL-M8812EU2, RTL8812EU, 5 GHz 2T2R	1	3	1,849	5,547	Hubtronics
Video antennas	6 dBi 5 GHz dipole + 165 mm U.FL lead	2	6	75	450	Robocraze
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	3	1,439	4,317	Robu
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	3	989	2,967	Robu
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3	3,939	11,817	Amazon India
BEC, video radio	Hobbywing 3 A UBEC, 5/6 V, 5 A peak, 2-6S	1	3	369	1,107	KitsGuru
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	3	279	837	Robu
Airframe, propulsion and power						**2,03,243**
Motors	Tarot TL96020, 5008, 340 KV	4	12	4,703	56,436	Robokits
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	4	16	1,641	26,256	Robokits

continued

Item	Part	/AC	Qty	Unit	Total	Source
Speed controllers	Hobbywing XRotor FPV G2, 4-in-1, 65 A, 3-6S	1	3	6,500	19,500	RC Mumbai
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	6	2,259	13,554	Robu
Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm	1	3	2,895	8,685	Indian Robo Store
Motor mounts	Tarot 25 mm motor mount, TL9602	4	12	1,119	13,428	Robu
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3	3,012	9,036	Amazon India
Landing gear	F450/F550 landing skid	1	3	549	1,647	Robu
Suspension springs	5 mm stainless double torsion spring	10	32	2	64	IndiaMART
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	–	1	949	949	Robu
Release servos	MG90S mini servo, 180 deg	4	12	149	1,788	Robocraze
Battery pack	GenX 22.8V 6S 16000mAh 25C/40C HV LiPo, AS150	1	3	12,197	36,591	Robokits
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3	3,000	9,000	Njour
Pack retention	300 mm LiPo strap, reusable	2	6	62	372	Robu
Main leads	10 AWG ultra-flexible silicone wire	–	1	149	149	Robu
Power connectors	Amass XT90S anti-spark, M/F pair	4	12	180	2,160	Robokits
Signal connectors	JST ZH 6-pin, 200 mm leads	1	3	23	69	Robu
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	3	1,139	3,417	Desertcart
Insulation	PVC heat-shrink sleeve, 150 mm	1	3	47	141	Robu
Flight-test support						**29,867**
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	–	1	16,519	16,519	zBotic
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	–	1	7,349	7,349	Robu
Cable management	Nylon zip ties, 350 mm, 100 pcs	–	1	135	135	Robu
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	–	1	159	159	Robu
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	–	1	560	560	Amazon India
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	–	1	145	145	Amazon India
Consumables	Fasteners, tape, connectors, filament	–	1	5,000	5,000	local
Ground station						**12,945**
Video receivers	LB-Link BL-M8812EU2, RTL8812EU, one per aircraft channel	–	3	1,849	5,547	Hubtronics
Receive panels	5.8 GHz 14 dBi directional panel, RP-SMA, 62x55 mm	–	3	566	1,698	FlyRobo
Panel pigtails	U.FL to RP-SMA female pigtail, RG178	–	3	99	297	IndiaMART
Receive antennas	6 dBi 5 GHz dipole + U.FL lead, second port	–	3	75	225	Robocraze
Receiver USB hub	Transcend HUB3 powered 4-port USB hub + adapter	–	1	2,653	2,653	Flipkart
Coordination base	EByte E22-900T22U, SX1262, USB	–	1	1,646	1,646	Hubtronics
Base station mount	Photographic tripod and adapter	–	1	879	879	Amazon India
Total parts, three aircraft						**4,93,978**

/AC = fitted per aircraft (– = bought once for the programme). *Qty* = programme quantity, including spares. Retail listings are GST-inclusive; GST is added only to quotation and B2B lines (14,961 of parts). Every source name links to the listing the price came from.

Battery selection

Each pack was run through the sizing model at its own mass, with the motors fixed. *Endurance* is hover time over mission time, and the policy needs ≥ 2.0. *Queue* is the 20 % reserve left after the third aircraft's 120 s hold plus two 53 s descents.

Pack	Mass, g	Wh	MTOW, kg	T/W	Endurance, ×	Queue, Wh
Li-ion 6S3P (previous)	1449	292	6.36	2.00	2.02	+1.1
LiPo 12 Ah	1424	266	6.34	2.01	1.85 **fail**	−3.7 **fail**
LiPo 13 Ah	∼1519	289	6.43	1.98	1.96 **fail**	−0.5 **fail**
LiPo 14 Ah	∼1636	311	6.55	1.94	2.06	+2.5
LiPo 16 Ah HV (chosen)	1880–2211	365	6.8–7.1	1.8–1.9	2.1–2.3	+5 to +9

Open before ordering

- **Battery: weigh it.** GenX publishes no mass; 1880–2211 g is bracketed from 165–194 Wh/kg. The cost of 16 Ah is thrust-to-weight: 1.8–1.9 on the same motors unless the stand shows ≥ 3.55 kgf per motor. Charge it as LiHV. Its AS150 plug needs AS150 mates, not XT90S. The 150 A fuse is now the tightest margin against a ∼136 A peak.
- **Failsafe parameters.** `params.py` and the pack model still describe the Li-ion pack. Once the LiPo arrives, measure its sag curve, then update both. Do not fly before that.
- **ESC.** The 65 A is a candidate. It is confirmed on the thrust stand for current, temperature and desync with 18 in props and 340 KV motors.
- **Base antenna cable.** 3 m of RG58, 50 Ω, TNC-male to SMA-male, made by a local RF cable shop. The INR 600 is an estimate, so get the quote before ordering.
- **Ground antennas.** Each receiver gets a 14 dBi panel aimed at its own aircraft, plus an omni on its second port: wfb-ng transmits from the ground too, so neither port can be left open. Aim the panels and confirm the polarisation, which the listing does not state. The listed range is for analog FPV, not wfb-ng. Coax lengths depend on the ground-station layout.
- **USB hub.** Check the adapter's rating when it arrives.

- **Confirmed:** the GenX F9P Air is dual-band (GPS L1/L2, Galileo E1/E5b, BDS B1/B2, GLONASS G1/G2), with a QMC5883 compass.